



Japan for Big Tech regulation

Japan to join the US and Europe to take on market abuses by the four Big Tech companies. <http://digit.in/nov20-11>



Walmart Canada buys 130 Teslas

Tesla (TSLA) has secured a massive order of 130 Tesla Semi electric trucks from Walmart Canada. <http://digit.in/nov20-12>



Marconi transmitting the first radio signal

transmitter and receiver in 1896 and was the first person to successfully transmit and receive long-range radio signals. The following year, Marconi sent the world's first-ever wireless communication over the sea where a message was transmitted over a distance of 6 kilometres. What did the message say, you ask? It simply said, "Are you ready". During 1894-1896, Jagadish Chandra Bose was the first to investigate millimetre wave communication when he reached an extremely high frequency of up to 60Hz during his experiments. At around the same time, German scientist Ferdinand Braun was also conducting similar experiments utilising an induction coil that was patented by Tesla. Both Marconi and Braun ended up winning the 1909 Nobel Prize for their accomplishments in wireless telegraphy. Nikola Tesla saw promise in wireless electricity but he failed to produce a halfway decent wireless power transmitter, so his efforts in the wireless technology field were veiled. However, in 1943, the US Supreme Court acknowledged Tesla's 1897 patent for a transmitter and receiver as a contributing factor to the invention of telegraphy and radio technology. The public use of radio began in 1907.

In 1927, the first-ever commercial radiotelephone service was set up which operated between Britain and

the US. Later, in 1946, Bell introduced the first American commercial mobile radio telephone service to private customers. The very next year, the humble transistor was invented by John Bardeen, William Shockley and Walter Brattain. This made it possible to marry communication and the computer, something that goes hand-in-hand today.

THE BIRTH OF WI-FI

Telstar, the first communication satellite was launched into orbit in 1962. Prior to this, microwave towers could flash TV shows and other information from one point to another through the air, which supplemented the landlines that had already spread across the globe. However, these signals reached their limit once they reached the ocean. Satellite transmission allowed for instant communication across the globe and also allowed for real-time international TV. In 1968, the ARPANET (Advanced Research Projects Agency Network) was developed by the Defense Advanced Research Projects Agency, and this is often heralded as the precursor of the modern internet.

Motorola researcher Martin Cooper made the world's first handheld cel-

lular call in 1973. He chose to call Joel S. Engel, his arch-rival and the head of AT&T cellular and said to him, "Joel, I'm calling you from a cellular phone, a real cellular phone, a hand-held, portable, real cellular phone." The device was, of course, brick-sized and weighed a whopping two and a half pounds. The battery life? About 30 minutes and it took 10 hours to charge as well!

Soon after, in 1979, the first commercially automated cellular network was launched by NTT (Nippon Telegraph and Telephone) in Japan, initially only in the metropolitan area of Tokyo. This was the 1G generation of cellular network, so to speak. For two decades, cellular technology saw incremental but important improvements. Phones got smaller and more affordable and network speeds increased considerably while the price reduced progressively. Soon after, the second generation of cellular connectivity came along and 2G was launched under the GSM standard in Finland in 1991. For the first time, calls could be encrypted and the wireless calls were a lot clearer with less static and noise. People could send SMS, picture messages, and MMS on their phones. This was nothing short of a cultural phenomenon which led to the mass adaptation of cellular devices by consumers. 2G transfer

speeds of 9.6 kbit/s (at the start of 2G) may seem paltry now, but they were almost a miracle back then.

RF CMOS is basically a radio frequency circuit that utilises mixed-signal (digital and analogue) MOS integrated circuit technology, fabricated on the CMOS process. It was developed in the late 1980s by Asad Abidi. This technology is widely used in modern wireless communications that doesn't just include cellular network but also Bluetooth, Wi-Fi, GPS receivers, broadcasting, vehicle communi-



A DynaTAC 8000X, the first commercially available portable, handheld cellphone (1983)